International Journal of Biological Macromolecules 151 (2020) 1240–1249

Contents lists available at ScienceDirect

International Journal of Biological Macromolecules

journal homepage: http://www.elsevier.com/locate/ijbiomac

Allosteric inhibition and kinetic characterization of *Klebsiella pneumoniae* CysE: An emerging drug target

Deepali Verma [a], Sunita Gupta [a], Rahul Saxena [a], Punit Kaur [b], Rachana R. [a], Sudha Srivastava [a], Vibha Gupta [a,*]

[a] Department of Biotechnology, Jaypee Institute of Information Technology, Noida 201309, India
[b] Department of Biophysics, All India Institute of Medical Sciences, New Delhi 110029, India

ARTICLE INFO

Article history:
Received 17 August 2019
Received in revised form 22 October 2019
Accepted 22 October 2019
Available online 18 November 2019

Keywords:
Klebsiella pneumoniae
CysE
Crystal structure
Allosteric inhibitor
Quercetin
Molecular dynamics simulations

ABSTRACT

The emergence and spread of multidrug-resistant strains of *Klebsiella pneumoniae* is a major concern that necessitates the development of unique therapeutics. The essential requirement of serine acetyltransferase (SAT/CysE) for survival of several human pathogens makes it a very promising target for inhibitor designing and drug discovery. In this study, as an initial step to structure-based drug discovery, CysE from *K. pneumonia* was structurally and biochemically characterized. Subsequently, blind docking of selected natural products into the X-ray crystallography determined 3D structure of the target was carried out. Experimental validation of the inhibitory potential of the top-scorers established quercetin as an uncompetitive inhibitor of Kpn CysE. Molecular dynamics simulations carried out to elucidate the binding mode of quercetin reveal that this small molecule binds at the trimer-trimer interface of hexameric CysE, a site physically distinct from the active site of the enzyme. Detailed analysis of conformational differences incurred in Kpn CysE structure on binding to quercetin provides mechanistic understanding of allosteric modulation. Binding of quercetin to CysE leads to conformation changes in the active site loops and proximal loops that affect its internal dynamics and consequently its affinity for substrate/cofactor binding, justifying the reduced enzyme activity.

© 2019 Elsevier B.V. All rights reserved.

1. Introduction

Klebsiella pneumoniae (Kpn) is an opportunistic pathogen, belonging to the Enterobacteriaceae family that causes disease when an individual has weakened immunity due to another disease. It mainly affects the immune-compromised individuals, chronic alcoholics and people suffering from diabetes mellitus and chronic obstructive pulmonary disease [1,2]. It is responsible for nosocomial infections (hospital-acquired infections), urinary tract infections (UTI), bloodstream infections and meningitis in neonatal and elderly. Emerging strains of this pathogen, that are resistant to almost all available antimicrobials are leading to ineffective antibiotic treatment worldwide. This antibiotic threat underscores the dire urgency to develop novel unexploited therapeutics against these hypervirulent/multidrug-resistant strains of Kpn [3,4]. For designing of novel drugs, identification and understanding of critical pathways/enzymes, whose disruption leads to the loss of pathogen's virulence is a prerequisite. In particular, the cysteine biosynthetic pathway is reported to be vital for growth, anti-oxidative defense mechanism and various cellular activities in *Entamoeba histolytica* (Ehi) [5] and *Salmonella typhimurium* (Sty) [6]. In this two-step pathway, CysE/SAT (serine acetyltransferase) transfers acetyl moiety from acetyl Coenzyme A (acetyl-CoA) to L-serine resulting in *O*-acetyl serine (OAS). Subsequently, pyridoxal 5′ phosphate-dependent *O*-acetyl serine sulfhydralase (OASS), also known as CysK, transfers reduced sulfide to OAS in a β-replacement reaction yielding L-cysteine [7]. Diverse molecular mechanisms exist for regulating this *de novo* cysteine biosynthetic pathway in bacteria [8]. CysE-CysK bi-enzyme complex formation for sensing the bacterial sulfur pool [9], feedback inhibition by L-cysteine and CysE's transient quaternary structure [10] are well documented processes at the enzymatic level.

Importance of CysE, the first enzyme of the pathway, is well established in the survival of *Mycobacterium tuberculosis* (Mtb) in its persistent phase [11–13]. A recent report on opportunistic pathogen *Serratia marcescens* illustrates additional roles of CysE in flagellar transport and phospholipase activity [14]. Furthermore, a *cysE* mutant of *E. coli* formed biofilm faster than the wild type suggesting the role in biofilm development [15]. Considering the importance of CysE in bacterial pathogenesis, targeting CysE of *K. pneumoniae* may potentiate the de-velopment of novel therapeutics against the rapidly emerging drug-resistant strains of this pathogen. Unexpectedly, although CysE is a well-characterized enzyme [16–20], only limited inhibitors of the

* Corresponding author: Department of Biotechnology, Jaypee Institute of Information Technology, A-10, Sector-62, Noida, Uttar Pradesh 201309, India.
E-mail addresses: rachana@jiit.ac.in (R. R.), sudha.srivastava@jiit.ac.in (S. Srivastava), vibha.gupta@jiit.ac.in (V. Gupta).

https://doi.org/10.1016/j.ijbiomac.2019.10.170
0141-8130/© 2019 Elsevier B.V. All rights reserved.

enzyme have been reported so far. Apart from feedback inhibitor L-cysteine [21], glycine is also reported as an active site inhibitor of *H. influenza* (Hin) CysE [22]. Potential inhibitors that blocked the proliferation of *E. histolytica* were discovered by virtual screening of National Cancer Institute (NCI) database against *E. coli* (Eco) CysE active site. The most potent compound, 3-Oxo-2-phenyl-3,5-dihydro-2H-pyrazolo (3,4-d) thieno (2,3-b)pyridine-7-carboxylic acid, exhibited greater than 50% inhibition of *E. coli* CysE at IC_{50} of 72 μM [23]. Another study has found out that *Staphylococcus aureus* (Sau) CysE can be inhibited with two natural compounds, namely, 11-oxo-ebracteolatanolide B and (4R,4aR)-dihydroxy-3-hydroxymethyl-7,7,10a-trimethyl-2,4,4a,5,6,6a, 7,8,9,10,10a,l0b dodecahydrophenanthro [3,2-b]furan-2-one) with a MIC value of 12.5 μg/ml and 25 μg/ml, respectively. These 2 compounds also inhibit the growth of methicillin-resistant *S. aureus* and biofilm formation [24].

The success or failure of a rational drug discovery program depends on the knowledge of the accurate 3D structure of a target protein. An accurate structure increases the chances of obtaining the valid hits from *in silico* screening of virtual libraries. X-ray crystallography is the most popular technique for determining protein structures. Fortuitously, due to the growing size of Protein Data Bank (PDB), the probability of availability of a good template for solving the phase problem by Molecular Replacement method is high. For CysE also, several crystal structures from all domains are available in the PDB that can serve as template(s) for generating a good initial model of Kpn CysE by X-ray crystallography approach. A comprehensive list of these structures along with pairwise identity with Kpn CysE is provided in the supplementary Table S1. *E. coli* CysE (PDB code: 1T3D) has 96% sequence identity with Kpn CysE and therefore using it as a search model is a straightforward choice. Complementing the static structural information of a ligand-bound receptor complex with Molecular Dynamics (MD) simulation, further enhances the mechanistic understanding behind the interactions. Prototypical examples have been presented in a recent review by Bouckaert and co-authors [25]. It is against this backdrop, the present study reports determination of crystal structure of Kpn CysE in complex with L-cysteine for input to *in silico* docking of natural compounds. Experimental verification of computationally predicted compounds and subsequent MD simulations identify a potent inhibitor that binds its target allosterically. In-depth investigation of the structural basis for allosteric inhibition points out to perturbation in the quaternary structure of the enzyme being responsible for the loss of activity.

2. Materials and methods

2.1. Materials

Hi-Media (India) was the supplier for *K. pneumoniae* genomic DNA, ampicillin, isopropyl -d-1-thiogalactopyranoside (IPTG) and quercetin, whereas, Fermentas (India) supplied Taq and Pfu polymerases, restriction enzymes and T4 DNA ligase. Amicon Ultra-15 (50 kDa), acetyl-CoA and L-serine were procured through Merck Millipore. Molecular weight protein markers namely, alcohol dehydrogenase, carbonic anhydrase, apoferritin, haemoglobin and natural products such as apocynin, berberine and mangiferin were purchased from Sigma-Aldrich (India) while catalase was procured from CDH (India). Ni-NTA agarose resin and vasicine were obtained from Qiagen and Natural Remedies (India), respectively.

2.2. Cloning of Kpn cysE

Cloning of Kpn *cysE* was carried out in *E. coli* expression vector pET21c (Novagen). Kpn *cysE* gene was amplified using Kpn genomic DNA with pfu:Taq polymerase in the ratio of 1:4 and gene specific forward and reverse primer (5′-GATTTAGCTAGCCATCACCATCACCATC ACCCGTGTGAAGAACTGGATATCG-3′) and (5′-GATTTAAAGCTTTTAAAT CCCGTCGCCGTACTCG-3′) respectively. The underlined sequences are restriction sites for NheI and HindIII, wherein Kpn *cysE* is cloned within the pET21c vector. The forward primer also includes the 6× histidine tag for ease in purification. Restriction digestion of vector and insert was carried out with above-stated enzymes and ligated using T4 DNA ligase. Screening of positive recombinant clones was carried out using colony PCR and double digestion with enzymes used for cloning. The positive clones were also validated by gene sequencing. All the steps for the construction of recombinant KcysE_pET21c were accomplished using the standard protocol of Sambrook and Russell [26].

2.3. Heterologous expression and affinity purification of Kpn CysE

Recombinant plasmid KcysE_pET21c was transformed into *E. coli* BL21(DE3) host and transformed colonies were cultured (at 37 °C, 200 rpm) in 1 L LB broth supplemented with ampicillin (100 μg/ml) till OD at 600 nm was 0.8. For induction of recombinant protein, 1 mM IPTG was added to the above culture and incubated at 30 °C with shaking for 3 hrs. Hereafter, all steps including harvesting, lysis, purification and concentration were strictly carried out at 4 °C. After pelleting the cells at 6000 rpm for 10 min, purification was initiated by resuspending the cell pellet in 200 ml lysis buffer (50 mM sodium phosphate buffer pH 8.0; 10 mM imidazole; 300 mM NaCl; 1 mM PMSF and 2 mM β-mercaptoethanol). The cells were lysed by sonication (10 sec ON, 10 sec OFF, 40% amplitude for 20 min) and the clear lysate was collected by centrifugation at 12000 rpm for 30 min and subjected to Ni-NTA affinity chromatography. The lysate was loaded on to 3 ml packed resin that was pre-equilibrated with the lysis buffer. Washing was carried out with 30 ml of the same buffer containing higher imidazole (30 mM). Recombinant Kpn CysE was eluted (in fractions of 1 ml) from the column with lysis buffer supplemented with 250 mM imidazole. After analyzing the purity and homogeneity of eluted fractions on 10% SDS PAGE and 7.5% native PAGE, fractions were pooled, dialyzed in storage buffer (20 mM Tris pH 8.0; 50 mM NaCl and 5% glycerol) and concentrated with amicon Ultra-15 50 kDa to ~30 mg/ml before preserving at −80 °C.

2.4. SAT activity

The activity of Kpn CysE was determined at 25 °C through SAT assay by using the protocol of Noji et al. [27] and as reported earlier [10]. The K_m and V_{max} of Kpn CysE for L-serine were calculated using Michalis Menten kinetics. For inhibition studies, 5 μM inhibitor (stock prepared in 100% ethanol but diluted with autoclaved milli-Q water for working solution), was added in the above reaction mix and monitored for enzyme activity.

2.5. Determination of oligomeric form of Kpn CysE

Size exclusion chromatography was performed to analyze the biologically relevant form of Kpn CysE. Analytical 4 ml S-300 Xtimate HPLC column (Welch Materials, Inc.) was pre-equilibrated with the storage buffer at a flow rate of 0.3 ml/min and 5 μl protein (0.1 mg/ml) was injected onto the column. The Column was calibrated with 5 known protein standards – apoferritin (440 kDa), catalase (240 kDa), alcohol dehydrogenase (150 kDa), haemoglobin (64 kDa) and carbonic anhydrase (29 kDa) for determining the oligomeric nature of the recombinant Kpn CysE.

2.6. Thermal denaturation and stability of Kpn CysE

CD spectra of recombinant Kpn CysE were acquired on spectropolarimeter JASCO J-815. Protein sample (0.3 mg/ml) in 20 mM sodium phosphate, pH 8.0; 50 mM NaCl and 5% glycerol buffer was passed through 0.2 μm filter and centrifuged at 12,000 rpm for 10 min. The data was collected between 200 and 260 nm in 1.0 cm quartz cuvette.

3 scans were recorded every 5 °C from 30 °C to 90 °C with a scan speed of 100 nm/min. The curve depicting the thermal denaturation of the enzyme was plotted between temperature and molar ellipticity (222 nm).

2.7. Crystallization of Kpn CysE in complex with L-cysteine

Purified Kpn CysE (5 mg/ml) and 2 mM L-cysteine was incubated for 30 min at 25 °C prior to setting up co-crystallization trials with JCSG Core suits I and II (Qiagen, India). The best crystals obtained were further optimized with the additive screen (Hampton Research, USA). Trials were set up at 293 K with crystallization drop comprising of 2 μl of protein solution (in a storage buffer), 1.6 μl of reservoir buffer and 0.4 μl additive for equilibration against 500 μl of 25% ethylene glycol as precipitant. Final diffraction grade crystals were achieved with 3% w/v trimethylamine N-oxide dehydrate additive.

2.8. Data collection and refinement

The 2.8 Å X-ray data on Kpn CysE co-crystallized with L-cysteine was collected at ESRF, BM14 beamline (Grenoble, France) at cryogenic temperature. The data was reduced, indexed, integrated and scaled using iMOSFLM program of CCP4 7.0 package [28]. *E. coli* CysE hexamer (PDB code: 1T3D) was used as a search model for Molecular Replacement solution with PHASER [29]. The solution yielded the initial structure with six molecules of Kpn CysE in the asymmetric unit. This structure was refined by REFMAC 5.8.0155 (with TLS) interspersed with several iterations of model building in COOT [30]. Phosphate, ethylene glycol and water molecules were built in the model during refinement. Structure factors and co-ordinates for Kpn CysE have been submitted to the PDB and have been assigned 6JVU PDB code.

2.9. Blind docking of natural compounds against Kpn CysE

Blind docking of the natural compounds was carried out against Kpn CysE hexameric structure using Autodock Vina [31]. This software needs input in pdbqt format and both receptor and ligand files were prepared through AutoDock Tools [32] that assigned polar hydrogens and partial Gastieger charges to the molecules. The grid size was set to 100 * 100 * 100 xyz points. The conformation with the lowest free energy of binding was extracted and visualized with receptor structure for interaction analysis using Pymol [33].

2.10. Molecular dynamics (MD) simulation of Kpn CysE with quercetin

MD simulation of all-atoms was performed on the hexameric Kpn CysE using Gromacs 5.0 package (ff99SB forcefield and TIP3P water model). Parameterization of quercetin was done by antechamber and GAFF forcefield. Acpype python script [34] was used to convert it into a Gromacs compatible format. The energy of the complex was minimized by steepest descent and conjugate gradient method with the tolerance of 1000 kJ mol^{-1} nm^{-1}. A short simulation (200 ps) was carried out with constant number volume and temperature at 300 K (NVT), with subsequent equilibration of 200 ps, where 1 bar of pressure was applied using Berendson's method (NPT). A production run of 50 ns was performed, with the timestep of 2 fs and trajectories were generated after every 100 ps. The production data were analyzed by Gromacs inbuilt program g_rmsd. The RMSD plots were generated using Xmgrace [35]. The binding free energy between receptor and quercetin was determined using molecular mechanics and Poisson-Boltzmann surface area (g_mmpbsa) tool [36]. It entailed computing the sum of ΔEMM (potential energy of molecular mechanics including both bonded and non-bonded interactions), ΔG_{psolv} (polar solvation free energy), ΔG_{npsolv} (nonpolar salvation free energy) and subtracting the TΔS (contribution of the entropy to the free energy). ΔG_{psolv} was determined by using the Poisson-Boltzmann equation whereas ΔG_{npsolv} was estimated by using the most widely used non-polar model i.e, solvent accessible surface area (SASA). The energy contributors were scored for 20 snapshots extracted at the interval of 0.5 ns from the production trajectories between 40 ns and 50 ns. Although data in the manuscript is representing results from one single MD run of 50 ns, independently repeated simulations on identical time scale reproduced the same effect.

2.11. Kpn CysE and quercetin binding kinetics

Interaction of quercetin and Kpn CysE was kinetically characterized by developing an electrochemical impedance-based sensor (detailed description of the methodology is beyond the scope and will be published elsewhere). Briefly, a biosensor was fabricated over a gold screen-printed electrode as reported earlier [37,38]. 5 μg of Kpn CysE protein (10 μl of 0.5 mg/ml) was immobilized over gold nanoparticles modified electrode by incubating for 15 min under ambient conditions. After protein immobilization, the current response of the sensor was recorded in 100 μl of electrolyte solution (0.5 mM ferri/ferrocyanide in 0.01 M PBS, pH 7.4). 10 μl of quercetin (10 μM) was added and the current was recorded again till it saturates. This was followed by washing the electrode with 100 μl of an electrolyte solution. The pH mediated dissociation of the enzyme-ligand complex was carried out by adding 10 μl of 1 M sodium citrate trihydrate buffer (pH 5.0) to the electrolyte solution. Protein-ligand interaction (K_D) was determined as the ratio of dissociation (k_{off}) and association (k_{on}) rates.

3. Results

3.1. Purification and Kinetic properties of Kpn CysE

Fig. 1A illustrates the localization of recombinant Kpn CysE in BL21 (DE3) as well as the profile of steps carried out during purification using Ni-NTA affinity chromatography. Fortunately, the soluble fraction of crude lysate has a large amount of recombinant protein (Mw ~ 30 kDa) (lane 4) facilitating the downstream purification process. On Coomassie-stained gel, the eluted fractions from Ni-NTA column (lanes 8–11) are 97% pure (quantified by ImageJ software). Further analysis on native gel shows one dominating band (96.5% population), indicating the homogenous nature of the purified enzyme (Fig. 1B). The total yield of Kpn CysE that can be obtained from 1 L of culture corresponds to ~ 30 mgs. The cleavage of the thio-ester bond from acetyl coenzyme A by Kpn CysE was measured spectrophotometrically at 232 nm. The kinetic parameters, K_m(ser) and V_{max} for Kpn CysE were calculated using Line-weaver Burk plot and are 84.1 μM and 30.3 μM/min, respectively (depicted by the filled squares in Fig. 2A). The specific activity, k_{cat} and catalytic efficiency of Kpn CysE turn out to be 60.2 ± 10.3 $\mu mol \cdot min^{-1} \cdot mg^{-1}$, 1818.0 ± 150.8 min^{-1} and 0.021 ± 0.003 $mM^{-1}min^{-1}$, respectively (Table 1).

3.2. Determination of oligomeric form of Kpn CysE

Analytical SEC profile in Fig. 3 reveals that although the highest peak corresponding to 190 kDa suggests hexameric nature of the protein, albeit the presence of peaks at 100 kDa and ~ 50 kDa indicate the existence of trimeric and monomeric species also. SEC advocates native Kpn CysE to exist majorly as a hexameric species but some population of lower oligomeric forms can also be observed.

3.3. Thermal denaturation and stability of Kpn CysE

The thermal stability of Kpn CysE was assessed with circular dichroism (CD) by the following variation in the CD spectra as a function of temperature, ranging from 30 °C to 90 °C (inset of Fig. 4). The plot between the increasing temperature and change in molar ellipticity (222 nm) reveals a sigmoidal curve (Fig. 4). It appears that the interactions contributing to quaternary assembly are more dominating

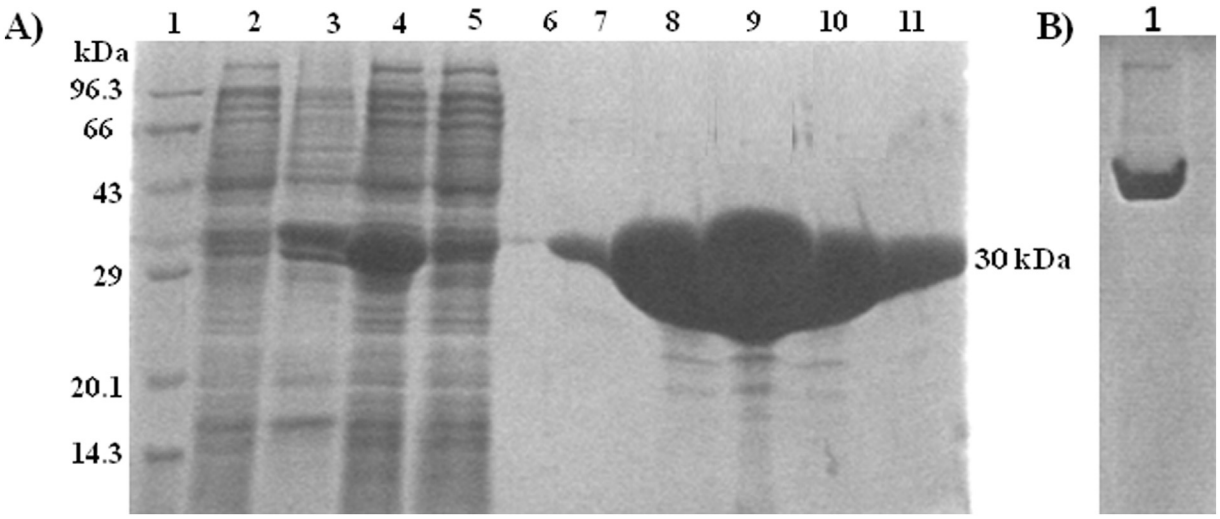


Fig. 1. Analysis of purification of Kpn CysE on (A) 10% SDS-PAGE: Lane 1, Protein Marker (in kDa), Lane 2, sample before induction; Lane 3, insoluble protein extract, Lane 4, soluble protein extract, Lane 5, flow through, Lane 6–7, wash with 20 mM and 30 mM imidazole, respectively, Lane 8 to 11, eluted fractions; (B) 7.5% native PAGE: Lane 1, homogenous and purified Kpn CysE.

than the interactions stabilizing the isolated monomer as no evidence is visible for stable folded monomers. The midpoint of the folding transition referred to as melting temperature T_m corresponds to 67.5 °C. The experimental data was not fitted to any model, as reversibility of unfolding by cooling the sample and then heating again could not be carried, owing to problems associated with the instrument availability.

3.4. Crystal structure of Kpn CysE in complex with L-cysteine

The data collection statistics for Kpn CysE-L-cysteine co-crystals and the refinement parameters for the final structure are listed in Table 2. The protein inhibitor complex crystallized in the monoclinic $P2_1$ space group with 6 protomers in the asymmetric unit. The model was refined to an R and R_{free} of 22.3% and 25.7%, respectively and consists of total 1517 residues, 2 phosphates, 1 ethylene glycol and 50 water molecules. Electron density for 5 residues at N-terminal and 14 residues at C-terminal was not traceable due to high disorder. The total accessible surface area of Kpn CysE calculated using PDBe PISA is 45903.4 Å^2 and the buried surface area is 21200.5 Å^2 [39]. The Gibbs free energy computed using the same software is 18.4 kcal/mol. The polypeptide chain folds into α-helix rich N-terminal domain (residues 5–140) and LβH C-terminal domain (residues 141–259). The structure of Kpn CysE is similar to Eco CysE, Hin CysE, *Brucella abortus* (Bab CysE), *Glycine max* (Gma) CysE and Ehi CysE with Root Mean Square Deviations (RMSD) of 0.252 Å, 0.479 Å, 0.685 Å, 0.867 Å and 0.742 Å, respectively (Fig. 5A). The substrate binding loop of Kpn CysE is ~ 10 residues long and its sequence is ^{184}GTGKTSGDRH^{193}. The small residues (especially Gly184 and Thr 185), Arg192 and His193 are well conserved in CysEs from all species. So is the length of the loop, exception being homologous CysEs from parasitic protozoa such as Entamoeba and Leishmania. This loop is longer due to insertion of eight amino acids and the cartoon structure of Ehi CysE (PDB code: 3P47) clearly illustrates this difference (Fig. 5A). As this substrate binding loop in most of the organisms is composed of small residues, it provided ample space for binding of substrate as well as inhibitor.

The 2.8 Å crystal structure of Kpn CysE supports the hexameric structure where two trimers are stacked on each other in head-to-head position through hydrophobic interactions at the N-terminus. A clear electron density is observed for cysteine molecule bound at the active site pocket at the interface of two Kpn CysE subunits (Fig. 5B). Each

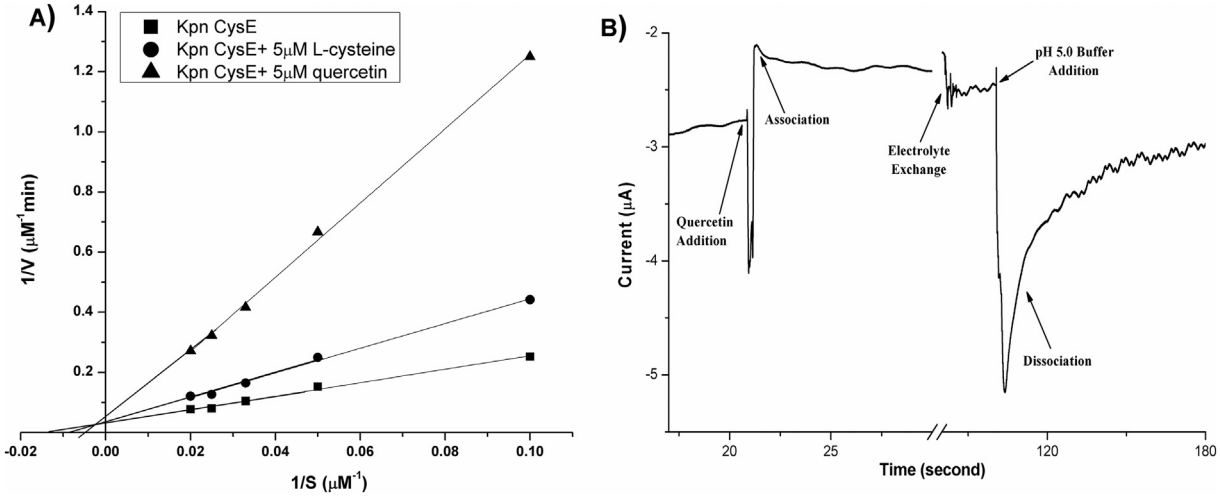


Fig. 2. (A) Kinetic parameters of Kpn CysE (square) and inhibition constant K_i of Kpn CysE with its feedback inhibitor L-cysteine (circle) and quercetin (triangle) was calculated using the Lineweaver-Burk plot. L-cysteine is a competitive inhibitor whereas quercetin displays a mixed type of inhibition of Kpn CysE; (B) Binding affinity between quercetin and Kpn CysE was determined using electrochemical technique. 5 μg of Kpn CysE was immobilized over electrode surface, association and dissociation of quercetin was monitored by variation in the output current. The association and dissociation rate $k_{on} = 2.55 \times 10^4$ μA s^{-1} M^{-1}, $k_{off} = 0.163$ μA s^{-1} and $K_D = 6.39$ μM were calculated for quercetin.

Table 1
Kinetic parameters for Kpn CysE.

Kinetic parameters	Kpn CysE
V_{max} (μM·min^{-1})	30.3 ± 5.2
K_m(ser) (μM)	84.1 ± 15.7
Specific activity (μmol·min^{-1}·mg^{-1})	60.2 ± 10.3
k_{cat} (min^{-1})	1818.0 ± 150.8
Catalytic efficiency (mM^{-1} min^{-1})	0.021 ± 0.003
K_i(Cys) (μM) 5 μM Cysteine	125.2 ± 17.2

Kpn CysE trimer in-houses three such cysteine molecules at the equivalent sites related by a 3-fold axis, that, in turn, are related to remaining 3 active sites by 2 fold symmetry. Cysteine at the interface of two subunits is stabilized by interactions from one subunit involving Asp92, Pro93 and Ala94 (residues of turn connecting α5 and α6 helices) and Asp157, His158 (residues of LβH loop) as well as from Gly183, Gly184, Thr185, Arg192, His193 (residues of substrate binding loop) of the ad-jacent subunit (Fig. 5C).

3.5. Blind docking of natural compounds in Kpn CysE

The crystal structure of hexameric Kpn CysE was used as a receptor for blind docking of quite a few natural compounds already recognized for their antibacterial, antioxidant, anti-inflammatory, bronco-dilatory and respiratory stimulant activities [40–44]. Blind docking of cysteine in Kpn CysE was used as a positive control for standardizing the docking parameters. Although L-cysteine docked into the active site with expected interactions (Figs. 5C and 6A), however, most of the natural compounds screened did not dock into the active site. The docking positions of some of the compounds namely, apocynin (Apo), berberine (Ber), mangiferin (Man), quercetin (Que) and vasicine (Vas) is depicted in Fig. 6A. The receptor in the figure is in grey cartoon representation whereas natural compounds as well as feedback inhibitor Cys are shown as grey sticks. It is apparent that Ber, Que and Vas are binding at the trimer-trimer interface (marked by a dashed black line in Fig. 6A). The VINA energy scores and receptor residues that are involved in H-bond formation or hydrophobic interaction with the ligands are listed in Table 3. All the selected compounds have higher docking scores than Cys but then they also have comparatively higher molecular weights. Que and Man have highest VINA energy scores (−8.3 and −8.2, respectively) and are involved in hydrophilic as well as hydrophobic interactions with Kpn CysE (Table 3).

3.6. Inhibition of Kpn CysE with natural compounds

The 5 top scorers (shortlisted in Table 3) were subsequently tested for their inhibitory potential against Kpn CysE by SAT assay. All the ligands (including cysteine) were assayed at 5 μM concentrations with total 0.5 μg of Kpn CysE enzyme in the reaction mix. Apocynin, berberine, mangiferin, quercetin and vasicine exhibited 41%, 24%, 22%, 62% and 46% inhibition of Kpn CysE, respectively (Fig. 6B). Out of these, quercetin exhibited the maximum inhibition (62%), better than its active site inhibitor L-cysteine (~50%), and appears to be a potent inhibitor of Kpn CysE (Fig. 6B). The IC_{50} value of this inhibitor, estimated through relative % inhibition, appears to be 3.7 μM.

3.7. MD simulation of Kpn CysE with quercetin

In order to confirm the binding position of quercetin in Kpn CysE, a 50 ns MD simulation was carried out on Kpn CysE complexed with quercetin as described in the material and methods sections. The backbone RMSD of proteins and Que show a well converged graph, indicating an equilibrated system (Fig. 6C). The total protein-ligand interaction energy (summation of short-range Coulombic interaction energy and Lennard-Jones energy) computed using 'gmx energy' yielded −246.22 ± 5.6 kJ/mol. The binding free energy calculated using g_mmpbsa is −131.346 ± 13.916 kJ/mol indicating favorable energetics. Superimposed pre (grey color) and post (salmon color) MD cartoon structures of the Kpn CysE receptor show movements throughout but noteworthy amongst these are α-3 helix, β-turn, α-4 helix and substrate binding LβH loop (Fig. 6A). In addition, after 50 ns of MD simulations, although the ligand Que has also moved (green stick), the binding position is still the interface of the two trimers. Binding of Que between the β-turns (Ala 54 - Pro 59 that joins α-3 and α-4) from three subunits at the trimer-trimer interface is stabilized by hydrogen bonds with Tyr 47, Asn 51 and Ile 57 as well as hydrophobic interactions with Ser 29, Phe 30, Pro 56, Ile 57, Met 58 and Ile 61 (Fig. 6D; Table 3). It appears that the reason behind the inhibition of Kpn CysE by Que lies in pushing away these β-turns and therefore displacing α-3 and α-4 helices and destabilizing the trimer-trimer interface.

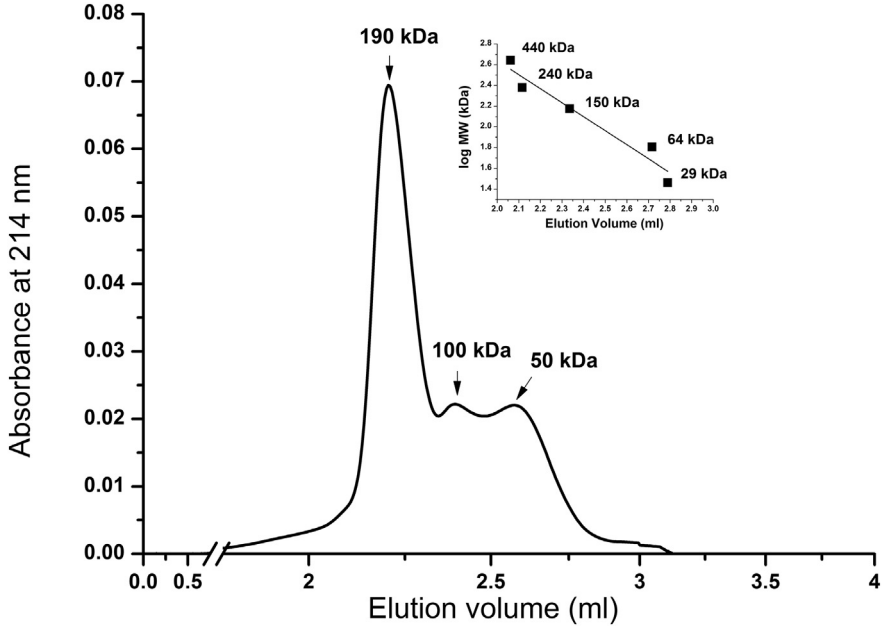


Fig. 3. SEC profile of Kpn CysE. The affinity purified protein was run under conditions described in Material and Methods section on analytical Xtimate S-300 column. The inset of the figure displays the standard plot of the column obtained with known proteins molecular weight markers.

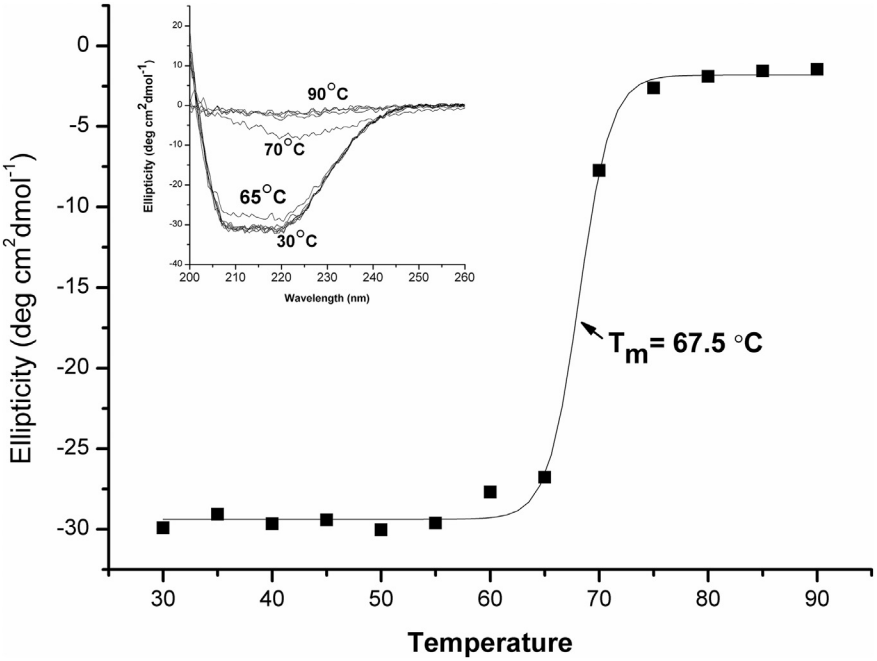


Fig. 4. Thermal denaturation of Kpn CysE. The T_m of Kpn CysE is 67.5 °C. Temperature-dependent CD spectra of Kpn CysE are illustrated in the inset.

3.8. Determination of K_i of quercetin for Kpn CysE

Kinetic parameters (K_i) and type of inhibition executed by quercetin on Kpn CysE were evaluated by the Lineweaver-Burk plot (Fig. 2A). The plot shows K_i of quercetin to be 162 μM for Kpn CysE (represented with triangles) and mixed type of inhibition with L-ser, since K_m is increased from 84.1 μM to 162 μM and V_{max} is decreased from 30.3 μM/min to 14.5 μM/min when compared to kinetic parameters of enzyme alone. As a positive control, L-Cys inhibits the enzyme competitively with an increase of K_m to 125.2 μM and no change in V_{max} values (depicted with circles) (Fig. 2A). These inhibitory assays corroborate with molecular docking, MD simulation analysis and establish that quercetin is an allosteric inhibitor of Kpn CysE that does not bind at active site and co-factor site rather it binds at trimer-trimer interface of hexameric CysE. Further, the affinity between quercetin and Kpn CysE receptor was evaluated in terms of dissociation constant (K_D = 6.39 μM) with the

Table 2
Data collection and refinement statistics.

Parameters	Value
Beamline	ESRF – BM14
Space Group	P 1 21 1
Unit-cell parameters (Å)	a = 68.74, b = 110.74, c = 101.74
Temperature (K)	100
Wavelength (Å)	0.9785
Resolution limit (Å)	74.66–2.8
Observed reflection	289,022 (42020)
Unique reflections	37,156 (5376)
Redundancy	7.8 (7.8)
Completeness (%)	99.3 (98.9)
Mean I/σ(I)	16.5 (2.1)
R_{merge}	0.067 (0.964)
No. of molecules in ASU	6
Matthews coefficient (Å^3 Da^{-1})	2.19
Solvent content (%)	43.81
Beamline	ESRF – BM14
Structure Refinement	
R_{free} test set	1766 reflections
R (%)	22.4
R_{free} (%)	25.69
Wilson B- factor (Å^2)	90.6
Average B, all atom (Å^2)	119
RMS deviations	
Bond length (Å)	0.015
Bond angles (°)	1.8
Ramachandran plot analysis	
Most favored/Allowed/Disallowed region	94.82%/5.11%/0.07%

Values in parenthesis represent the calculated data in highest resolution shell. R_{free} was computed with 5% of the data omitted from the refinement.

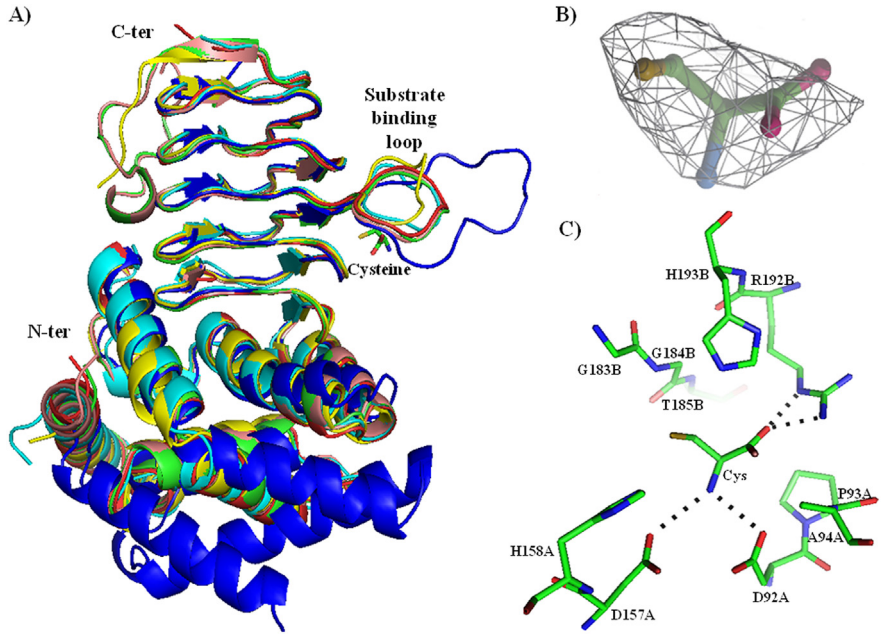


Fig. 5. (A) Structural alignment of Kpn CysE monomer (chain A, green) with the monomers of Eco CysE (1T3D:A, salmon), Hin CysE (1SSQ:A, magenta), Bab CysE (4HZC:A, yellow), Gma CysE (4N69:A, cyan) and Ehi CysE (3P47:A, Blue); (B) $2F_o$-F_c electron density map at 1.0 sigma level displaying L-cysteine bound in the active site of Kpn CysE; (C) The active site architecture of Kpn CysE represents the interaction of L-cysteine with the active site residue and the dotted line shows the interaction via hydrogen bonds.

association (k_{on}) and dissociation (k_{off}) rates as 2.55×10^4 μA s^{-1} M^{-1} and 0.163 μA s^{-1}, respectively (as shown in Fig. 2B).

4. Discussions

Kpn is a notorious pathogen that causes nosocomial infections and UTI in person with a weaker immune system. The emergence of the antibiotic-resistant strains of *K. pneumoniae*, has necessitated exploration for novel inhibitors against essential targets to develop effective therapeutics against these strains [1,4]. CysE, a gate enzyme for the cysteine biosynthetic pathway, is one such target [5,6,11]. This work contributes to the first phase of structure-based drug discovery process by reporting the recombinant production, structural and biochemical characterization of CysE from *K. pneumoniae*. The overall structure of Kpn CysE, as expected, is very similar to homologous CysEs (Fig. 5A) and the feedback inhibitor Cysteine makes interaction with the active site residues as reported (Fig. 5C). SEC carried out for determination of the biologically relevant form of Kpn CysE shows that the dominating population is hexameric which is interspersed with a small fraction of trimeric and monomeric species (Fig. 3). CysE from Eco [16], Hin [17] and Gma [20] are established hexamers but a recent study has demonstrated similar heterogeneous ensemble for *E. coli* / *S. flexneri* CysE as well. It is further proposed that the existence of these minor forms could be concentration-dependent or time-dependent and may perhaps be another mechanism for regulating cysteine production [10]. A CysE homolog from Bab is reported to exist both as a trimer and a hexamer [19] whereas Ehi CysE is categorically a trimer due to the different orientation of N-terminal helices, the kink between α-4 and α-5 helix and an additional insertion of 7 residues in the middle of N terminal domain [18]. The inability to form hexamer affects the CSC formation and hence eliminates this regulatory mechanism, leaving cysteine feedback inhibition the only regulatory mechanism for *de novo* cysteine production in *E. histolytica* and ensuring sufficient cysteine availability for defense against oxidative stress [18]. Likewise, a single mutation (A241V) in a natural variant of Sfl CysE results in loss of hexameric quaternary structure leading to reduced activity and stability. This work suggested that disturbing the hexameric assembly of CysE may be another mechanism for regulating intracellular levels of cysteine [10]. In the stability context, it is useful to point out that T_m of hexameric Kpn CysE (67.5 °C; Fig. 4) is similar to T_m of 65 °C reported for homologous Eco CysE [10]. Comparative analysis of specific activities as reported for Kpn CysE homologs reveal that it is comparable to 71.6–-106 μmol/min/mg specific activity reported for Eco CysE [10,45] as well as 10.66 μmol/min/mg for Mtb CysE [46] but much higher than 0.0038 μmol/min/mg reported for Sau CysE [24]. Comparison of K_m for CysEs from different organisms (Table S2), indicates that the Kpn CysE has lowest K_m (84.1 μM) and therefore higher affinity for L-serine than Eco CysE (145 μM), Bab CysE (127.5 μM), Gma CysE 617 μM, Hin CysE (18 mM), Sty CysE (560 μM) and Sau CysE (458.7 μM) [10,19–22, 24]. The inhibition kinetics of Kpn CysE with its feedback inhibitor L-cysteine shows that it inhibits Kpn CysE competitively like other CysEs from Eco CysE [16], Hin CysE [47], Ehi CysE [18] and Bab CysE [19] (Fig. 2A).

The 3D crystal structure of L-cysteine bound Kpn CysE formed the basis for blind docking of several natural compounds within the Kpn CysE structure. Activity analysis supported the *in silico* hypothesis where several ligands such as apocynin, berberine, mangiferin, quercetin and vasicine reduced Kpn CysE activity by 22–62% with quercetin displaying the maximum inhibition at 5 μM concentration (Fig. 6B). Detailed kinetic experiments with quercetin suggests it to be a mixed type of inhibitor of Kpn CysE, since its binding increases the K_m but decreases the V_{max} of the enzyme (Fig. 2A).

In view of the fact that the docked poses indicted binding sites for most of these natural compounds to be non-overlapping with substrate or cofactor binding sites (Fig. 6A), MD approach was undertaken for understanding the mechanistic basis of inhibition. Multiple independent MD simulation runs provide confidence in declaring that quercetin binds and disturbs the trimer-trimer interface of Kpn CysE (Fig. 6C) Binding of quercetin causes the repulsive movement of β-turns (that connects α-3 and α-4 helices) from the two trimers as well as conformational changes in the substrate binding LβH loop (Fig. 6A). Interestingly, an earlier study with Sfl A241V CysE variant clearly

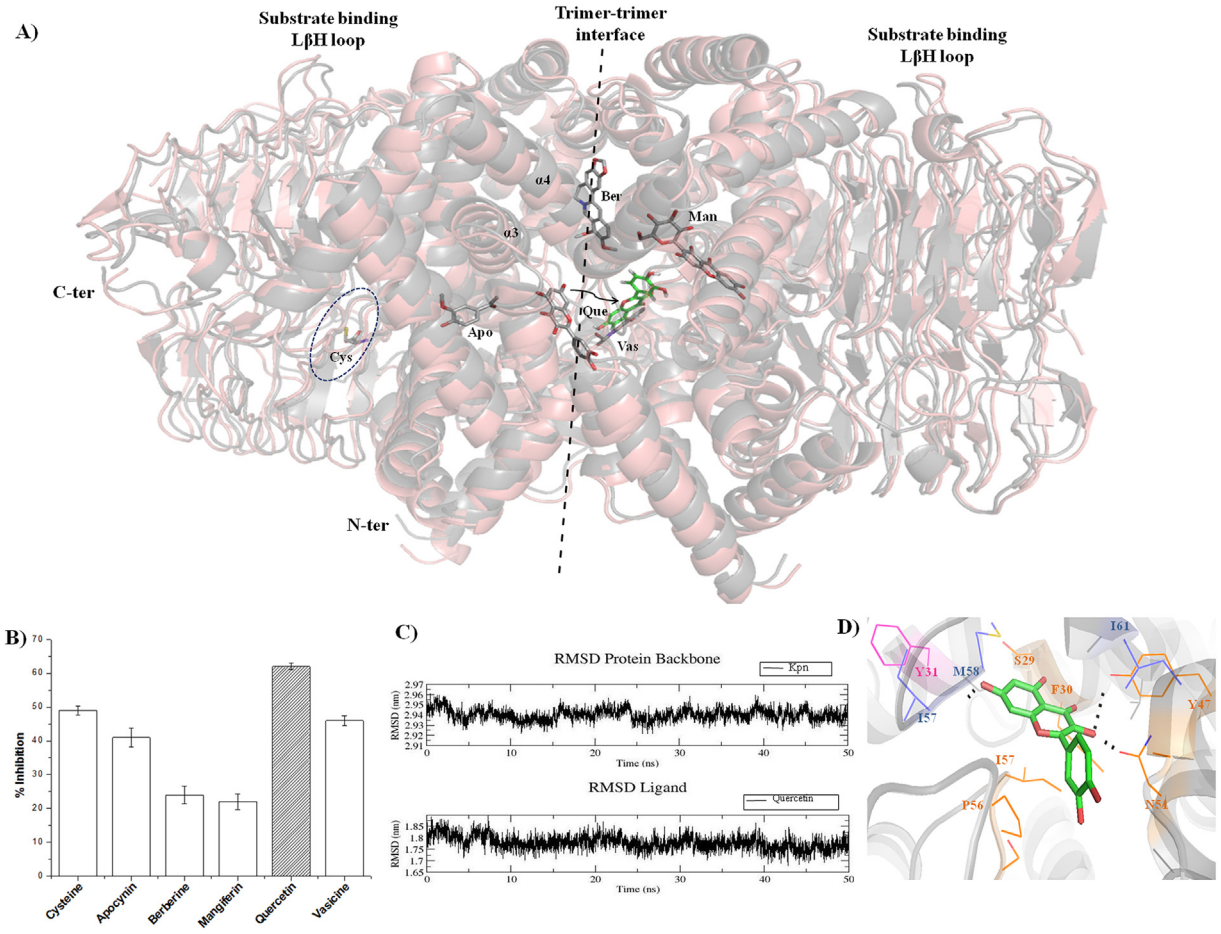


Fig. 6. (A) Cartoon representation of superimposed hexameric Kpn CysE structures before MD (grey) and after 50 ns of MD simulation (salmon). The location of the active site and the trimer-trimer interface is marked with a dashed oval and dashed line, respectively. The docked natural compounds and cysteine are shown in grey sticks where cysteine is denoted by Cys, apocynin by Apo, berberine by Ber, and mangiferin by Man, quercetin by Que and vasicine by Vas; (B) % inhibition of Kpn CysE with its feedback inhibitor L-cysteine and natural compounds- apocynin, berberine, mangiferin, quercetin and vasicine; (C) Backbone RMSD versus simulation time for 50 ns MD simulations of Kpn CysE hexamer (receptor) and quercetin (ligand); (D) The magnified view of quercetin binding site after 50 ns of MD simulations where quercetin interacts with the residues represented in sticks and dotted line depicts the interaction via hydrogen bonds. The residues from different subunits are color coded in orange, magenta and blue.

demonstrated movements of the same structural elements, which led to disso-ciation of quaternary assembly of Sfl CysE, loss of enzyme activity and structural stability [10]. Therefore, it stands to reason, that binding of quercetin within Kpn CysE is causing subtle structural distortions that are modulating the quaternary assembly and substrate/cofactor binding sites, thus reducing its affinity for L-serine/acetyl-CoA. The binding affinity between the enzyme and the inhibitor was determined electrochemically and analysis of the rates of forward ($k_{on} = 2.55 \times 10^4$ μA s^{-1} M^{-1}) and reverse reactions ($k_{off} = 0.163$ μA s^{-1}) yielded K_D of 6.39 μM indicating reasonably high affinity between them (Fig. 2B). Significantly lower k_{off} compared to k_{on} indicates that quercetin is bound to Kpn CysE for an extended period of time which is favorable for its inhibitory action.

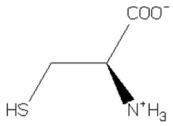
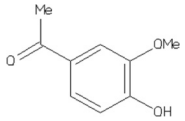
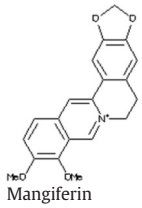
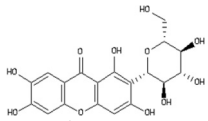
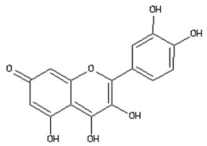
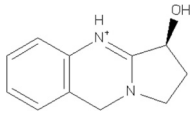
Since the success of benzodiazepines, allosteric modulators of the neurotransmitter γ-aminobutyric acid (GABA) over direct-acting $GABA_A$ agonists, considerable focus is being placed in allosteric drug discovery for minimizing undesirable side effects [48]. Similarly, novel allosteric drugs have been developed for G-protein-coupled receptors (GPCRs) that offer high selectivity and used in the treatment of multiple psychiatric and neurological human disorders [49]. Another example of allosteric inhibitor is sulfated benzofurans that inhibit the action of thrombin, (an enzyme that acts as coagulating factor) also fine-tune its regulation and evolved as a new class of therapeutics against thrombin [50].

Quercetin is reported to allosterically inhibit an enzyme of glyoxylate shunt pathway namely, isocitrate lyase (ICL) *of M. tuberculosis*, by binding to its N-terminus. There too it exhibits the noncompetitive inhibition with an IC_{50} value of 3.57 μM [51]. It also inhibits the ICL of *Candida albicans* [52] and DNA gyrase of mycobacterial species [53]. Further, it is reported that administrating TB drugs with quercetin and polyvinylpyrrolidone, accelerated the treatment period and reduced the adverse effects of the antimycobacterial drugs [54]. This study demonstrates that quercetin has another cellular target, namely CysE, advo-cating clinical advantage of quercetin in being developed as a lead compound against bacterial infections.

5. Conclusions

In conclusion, structural and functional characterization of Kpn CysE along with the search for natural product inhibitor has concluded in identification of quercetin as an "allosteric inhibitor" and a potential lead candidate against CysE from *Klebsiella pneumonia*. This study provides the conclusive evidence that quercetin inhibits Kpn CysE by binding allosterically at the trimer-trimer interface of hexmaric CysE and not at the active site *per se*. Where competitive inhibitors have not proven fruitful for bacterial pathogens utilizing the cysteine biosynthetic pathway, identification of an allosteric inhibitor of CysE has biological importance in allowing modulation of the enzyme activity. In addition, as

Table 3
Molecular docking results of natural products inhibitor against Kpn CysE using Auto dock VINA.

S. No.	Compound name & 2D structure	Mol. Wt. (g/mol)	Vina Energy Score (kcal/mol)	H bonds with receptor	Hydrophobic interactions with receptor
1	Cysteine 	121.16	−4.3	(N)-(O1) Asp92A, (N)-(O2) Asp157A, (O)-(NH1) Arg192B, (O)-(NE) Arg192B	Pro93A, Ala94A, His158A, Gly183B, Gly184B, Thr185B, His193B
2	Apocynin 	166.17	−5.5	(O)-(NH2) Gln135B	Leu103A, Tyr104A, Lys106A, His109A, Phe140A, Ser55B, Pro56B, Ile57B, Ile130B Phe131B, Asn134B, Val138B
3	Berberine 	336.36	−8.6	_	Pro56A, Ile57A, Ser29C, Phe30C, Tyr47C, Ala50C, Asn51C, Lys52C, Ala54C, Ile61C, Ala54E, Ser55E, Pro56E, Ile57E, Met58E, Pro59E, Ile61E
4	Mangiferin 	422.34	−8.2	(O5)-(NH2) Asn51D, (O4)-(NH3) Lys52D, (O9)-(NH2) Gln141E, (O8)-NH1 Gln141F	Ser139D, Phe140D, Gln141D, Ser55E, Pro56E, Ile57E, Gln135E, Ser139E, Phe140E,
5	Quercetin 	302.23	−8.3	(O4)-(H1) Tyr47A, (O4)-(NH2) Asn51A, (H1)-(O) Ile57D	Ser29A, Phe30A, Pro56B, Ile57B, Met58B, Pro59B, Ser29D, Tyr31D Pro56D, Pro59D, Ile61D
6	Vasicine (Peganine) 	188.23	−7.0	(H1)-(O) Met26B	Met26B, Leu27B, Ala28B, Ser29B, Phe30B, Thr34E, Ser29E, Phe30E, Tyr47E, Asn51E, Pro56F, Ile57F, Met58F, Pro59F

limited compounds have been documented till date for CysE inhibition, quercetin and other inhibitors reported in this study are additions to the scanty inhibitor database of the enzyme. This work also ascertains that trimer-trimer interface of the hexameric CysE is an internally regulated allosteric site and should be exploited further for understanding and modulating functional dynamics.

Author contributions

DV carried out the cloning, purification, biophysical and biochemical characterization of Kpn CysE and wrote the manuscript; SG performed docking, molecular dynamics simulations and analysis; RS carried out experiments for determining binding affinity between quercetin and Kpn CysE; PK helped in analyzing the Kpn CysE structure and critical evaluation of the manuscript; RR contributed all the natural compounds; SS provided resources, supervised and interpreted the receptor-inhibitor binding affinity data; VG conceived, planned, analyzed data, helped in manuscript preparation and supervised the study.

Acknowledgements

The authors are indebted to the Department of Biotechnology (DBT), Govt. of India (GOI) for research grant [Ref No. BT/BioCARe/01/79/2011-12]. We gratefully acknowledge the team of European Synchrotron Radiation Facility (ESRF) Beamline 14 and DBT, GOI for providing access to the beamline.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2019.10.170.

References

[1] M.K. Paczosa, J. Mecsas, Klebsiella pneumoniae: going on the Offense with a Strong Defense, Microbiol. Mol. Biol. Rev. 80 (2016) 629–661, https://doi.org/10.1128/MMBR.00078-15.
[2] Klebsiella pneumoniae in Healthcare Settings, (n.d.). <https://www.cdc.gov/hai/organisms/klebsiella/klebsiella.html> (accessed April 13, 2019).
[3] R. Podschun, U. Ullmann, Klebsiella spp. as Nosocomial Pathogens: epidemiology, taxonomy, typing methods, and pathogenicity factors, Clin. Microbiol. Rev. 11 (1998) 589–603. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC88898/pdf/cm000589.pdf> (accessed April 13, 2019).
[4] L. Diancourt, V. Passet, J. Verhoef, P.A.D. Grimont, S. Brisse, Multilocus Sequence Typing of Klebsiella pneumoniae Nosocomial Isolates, J. Clin. Microbiol. 43 (2005) 4178–4182, https://doi.org/10.1128/JCM.43.8.4178-4182.2005.
[5] F.D. Gillin, L.S. Diamond, Attachment of Entamoeba histolytica to glass in a defined maintenance medium: specific requirement for cysteine and ascorbic acid, J. Protozool. 27 (1980) 474–478, (accessed June 6, 2019) http://www.ncbi.nlm.nih.gov/pubmed/6260930.
[6] A.L. Turnbull, M.G. Surette, Cysteine biosynthesis, oxidative stress and antibiotic resistance in Salmonella typhimurium, Res. Microbiol. 161 (2010) 643–650, https://doi.org/10.1016/j.resmic.2010.06.004.

[7] T. Leustek, M.N. Martin, J.-A. Bick, J.P. Davies, Pathways and regulation of sulfur metabolism revealed through molecular and genetic studies, Annu. Rev. Plant Physiol. Plant Mol. Biol. 51 (2000) 141–165, https://doi.org/10.1146/annurev.arplant.51.1.141.

[8] Eric Guédon, Isabelle Martin-Verstraete, Microbiology MonographsAmino Acid Biosynthesis ~ Pathways, Regulation and Metabolic Engineering, Springer Berlin Heidelberg, Berlin, Heidelberg 2007, pp. 195–218, https://doi.org/10.1007/7171_2006_060.

[9] J.M. Jez, S. Dey, The cysteine regulatory complex from plants and microbes: what was old is new again, Curr. Opin. Struct. Biol. 23 (2013) 302–310, https://doi.org/10.1016/j.sbi.2013.02.011.

[10] D. Verma, S. Gupta, K.J. Kaur, V. Gupta, Is perturbation in the quaternary structure of bacterial CysE, another regulatory mechanism for cysteine synthesis? Int. J. Biol. Macromol. 111 (2018) 1010–1018, https://doi.org/10.1016/J.IJBIOMAC.2018.01.076.

[11] C.M. Sassetti, D.H. Boyd, E.J. Rubin, Genes required for mycobacterial growth defined by high density mutagenesis, Mol. Microbiol. 48 (2003) 77–84, (accessed September 30, 2018) http://www.ncbi.nlm.nih.gov/pubmed/12657046.

[12] C.M. Sassetti, E.J. Rubin, Genetic requirements for mycobacterial survival during infection, Proc. Natl. Acad. Sci. 100 (2003) 12989–12994, https://doi.org/10.1073/pnas.2134250100.

[13] J. Rengarajan, B.R. Bloom, E.J. Rubin, From The Cover: genome-wide requirements for Mycobacterium tuberculosis adaptation and survival in macrophages, Proc. Natl. Acad. Sci. 102 (2005) 8327–8332, https://doi.org/10.1073/pnas.0503272102.

[14] M.T. Anderson, L.A. Mitchell, H.L.T. Mobley, Cysteine Biosynthesis Controls Serratia marcescens phospholipase activity, J. Bacteriol. 199 (2017) e00159–e217, (accessed June 8, 2019) http://jb.asm.org/.

[15] G. Sturgill, C.M. Toutain, J. Komperda, G.A. O'Toole, P.N. Rather, Role of CysE in production of an extracellular signaling molecule in Providencia stuartii and Escherichia coli: Loss of cysE enhances biofilm formation in Escherichia coli, J. Bacteriol. 186 (2004) 7610–7617, https://doi.org/10.1128/JB.186.22.7610-7617.2004.

[16] V.E. Pye, A.P. Tingey, R.L. Robson, P.C.E. Moody, The structure and mechanism of serine Acetyltransferase from *Escherichia coli*, J. Biol. Chem. 279 (2004) 40729–40736, https://doi.org/10.1074/jbc.M403751200.

[17] J. Gorman, L. Shapiro, Structure of serine acetyltransferase from *Haemophilus influenzae* Rd, Acta Crystallogr. D Biol. Crystallogr. 60 (2004) 1600–1605, https://doi.org/10.1107/S0907444904015240.

[18] S. Kumar, I. Raj, I. Nagpal, N. Subbarao, S. Gourinath, Structural and Biochemical Studies of Serine Acetyltransferase Reveal Why the Parasite *Entamoeba histolytica* Cannot Form a Cysteine Synthase Complex, J. Biol. Chem. 286 (2011) 12533–12541, https://doi.org/10.1074/jbc.M110.197376.

[19] S. Kumar, N. Kumar, N. Alam, S. Gourinath, Crystal structure of serine acetyl transferase from Brucella abortus and its complex with coenzyme A, Biochimica et Biophysica, Acta (BBA) – Prote. Proteom. 2014 (1844) 1741–1748, https://doi.org/10.1016/j.bbapap.2014.07.009.

[20] H. Yi, S. Dey, S. Kumaran, S.G. Lee, H.B. Krishnan, J.M. Jez, Structure of soybean serine acetyltransferase and formation of the cysteine regulatory complex as a molecular chaperone, J. Biol. Chem. 288 (2013) 36463–36472, https://doi.org/10.1074/jbc.M113.527143.

[21] N.M. Kredich, G.M. Tomkins, The Enzymic Synthesis of L-Cysteine in Escherichia coli and Salmonella typhimurium, J. Biol. Chem. 241 (1966) 4955–4966, (accessed November 20, 2017) http://www.jbc.org/content/241/21/4955.full.pdf.

[22] C.M. Johnson, B. Huang, S.L. Roderick, P.F. Cook, Kinetic mechanism of the serine acetyltransferase from Haemophilus influenzae, Arch. Biochem. Biophys. 429 (2004) 115–122. doi:10.1016/j.abb.2004.06.006.

[23] S. Agarwal, R. Jain, A. Bhattacharya, A. Azam, Inhibitors of Escherichia coli serine acetyltransferase block proliferation of Entamoeba histolytica trophozoites, Int. J. Parasitol. 38 (2008) 137–141, https://doi.org/10.1016/j.ijpara.2007.09.009.

[24] C. Chen, Q. Yan, M. Tao, H. Shi, X. Han, L. Jia, Y. Huang, L. Zhao, C. Wang, X. Ma, Y. Ma, Characterization of serine acetyltransferase (CysE) from methicillin-resistant Staphylococcus aureus and inhibitory effect of two natural products on CysE, Microb. Pathog. 131 (2019) 218–226, https://doi.org/10.1016/j.micpath.2019.04.002.

[25] J. De Ruyck, G. Roos, E.M. Krammer, M. Prévost, M.F. Lensink, J. Bouckaert, Chapter 4: Molecular Mechanisms of Drug Action: X-ray Crystallography at the Basis of Structure-based and Ligand-based Drug Design, in: RSC Drug Discovery Series, Royal Soc. Chem., (2018) 67–86. doi:10.1039/9781788010016-00067.

[26] J. Sambrook, D.W. (David W. Russell, Molecular cloning : a laboratory manual, Cold Spring Harbor Laboratory Press, 2001.

[27] M. Noji, K. Inoue, N. Kimura, A. Gouda, K. Saito, Isoform-dependent differences in feedback regulation and subcellular localization of serine acetyltransferase involved in cysteine biosynthesis from Arabidopsis thaliana, J. Biol. Chem. 273 (1998) 32739–32745, https://doi.org/10.1074/JBC.273.49.32739.

[28] T.G.G. Battye, L. Kontogiannis, O. Johnson, H.R. Powell, A.G.W. Leslie, *iMOSFLM* : a new graphical interface for diffraction-image processing with *MOSFLM*, Acta Crystallogr. D Biol. Crystallogr. 67 (2011) 271–281, https://doi.org/10.1107/S0907444910048675.

[29] A.J. McCoy, R.W. Grosse-Kunstleve, P.D. Adams, M.D. Winn, L.C. Storoni, R.J. Read, *Phaser* crystallographic software, J. Appl. Crystallogr. 40 (2007) 658–674, https://doi.org/10.1107/S0021889807021206.

[30] P. Emsley, B. Lohkamp, W.G. Scott, K. Cowtan, Features and development of Coot, Acta Crystallogr. Sect. D Biol. Crystallogr. D 66 (2010) 486–501, https://doi.org/10.1107/S0907444910007493.

[31] O. Trott, A.J. Olson, AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading, J. Comput. Chem. 31 (2010) 455–461, https://doi.org/10.1002/jcc.21334.

[32] G.M. Morris, R. Huey, W. Lindstrom, M.F. Sanner, R.K. Belew, D.S. Goodsell, A.J. Olson, AutoDock4 and AutoDockTools4: automated docking with selective receptor flexibility, J. Comput. Chem. 30 (2009) 2785–2791, https://doi.org/10.1002/jcc.21256.

[33] Warren DeLano, The PyMOL Molecular Graphics System, (n.d.). <http://pymol.sourceforge.net/overview/index.htm> (accessed April 27, 2019).

[34] A.W. Sousa da Silva, W.F. Vranken, ACPYPE - AnteChamber PYthon Parser interfacE, BMC Res. Notes 5 (2012) 367, https://doi.org/10.1186/1756-0500-5-367.

[35] P. Turner, XMGRACE, Version 5.1. 19, Center for Coastal and Land-Margin Research, Oregon, 2005.

[36] R. Kumari, R. Kumar, A. Lynn, A. Lynn, *g_mmpbsa* —A GROMACS Tool for High-Throughput MM-PBSA Calculations, J. Chem. Inf. Model. 54 (2014) 1951–1962, https://doi.org/10.1021/ci500020m.

[37] R. Saxena, S. Srivastava, Nanoparticles empowered microelectrode for fast and sensitive detection of thyroid stimulating hormone, Sens. Lett. 15 (2017) 375–379, https://doi.org/10.1166/sl.2017.3831.

[38] R. Raghav, S. Srivastava, Immobilization strategy for enhancing sensitivity of immunosensors: L -Asparagine–AuNPs as a promising alternative of EDC–NHS activated citrate–AuNPs for antibody immobilization, Biosens. Bioelectron. 78 (2016) 396–403, https://doi.org/10.1016/j.bios.2015.11.066.

[39] E. Krissinel, K. Henrick, Inference of macromolecular assemblies from crystalline state, J. Mol. Biol. 372 (2007) 774–797.

[40] A.V. Anand David, R. Arulmoli, S. Parasuraman, Overviews of biological importance of quercetin: a bioactive flavonoid, Pharmacogn. Rev. 10 (2016) 84–89, https://doi.org/10.4103/0973-7847.194044.

[41] J. Stefanska, R. Pawliczak, Apocynin: molecular aptitudes, Mediat. Inflamm. 2008 (2008), 106507. https://doi.org/10.1155/2008/106507.

[42] K. Nepali, S. Sharma, R. Ojha, K.L. Dhar, Vasicine and structurally related quinazolines, Med. Chem. Res. 22 (2013) 1–15, https://doi.org/10.1007/s00044-012-0002-5.

[43] A. Vyas, K. Syeda, A. Ahmad, S. Padhye, F.H. Sarkar, Perspectives on medicinal properties of Mangiferin, Mini-Rev. Med. Chem. 12 (2012) 412–425, https://doi.org/10.2174/138955712800493870.

[44] Micol Tillhon, Luis M. Guamán Ortiz, Paolo Lombardi, A. Ivana Scovassi, Berberine: new perspectives for old remedies, Biochem. Pharmacol. 84 (10) (2012) 1260–1267, https://doi.org/10.1016/j.bcp.2012.07.018.

[45] D.B. Wigley, J.P. Derrick, W. V Shaw, The serine acetyltransferase from Escherichia coli. Over-expression, purification and preliminary crystallographic analysis, FEBS Letters. 277 (1990) 267–71. <http://www.ncbi.nlm.nih.gov/pubmed/2125278> (accessed June 13, 2019).

[46] J. Qiu, D. Wang, Y. Ma, T. Jiang, Y. Xin, Identification and characterization of serine acetyltransferase encoded by the Mycobacterium tuberculosis Rv2335 gene, Int. J. Mol. Med. 31 (2013) 1229–1233, https://doi.org/10.3892/ijmm.2013.1298.

[47] L.R. Olsen, B. Huang, M.W. Vetting, S.L. Roderick, Structure of serine acetyltransferase in complexes with CoA and its cysteine feedback inhibitor, Biochemistry 43 (2004) 6013–6019, https://doi.org/10.1021/bi0358521.

[48] H. Mohler, J.M. Fritschy, U. Rudolph, A new benzodiazepine pharmacology, J. Pharmacol. Exp. Ther. 300 (2002) 2–8, https://doi.org/10.1124/jpet.300.1.2.

[49] P. Jeffrey Conn, A. Christopoulos, C.W. Lindsley, Allosteric modulators of GPCRs: a novel approach for the treatment of CNS disorders, Nat. Rev. Drug Discov. 8 (2009) 41–54, https://doi.org/10.1038/nrd2760.

[50] P.S. Sidhu, A. Liang, A.Y. Mehta, M.H. Abdel Aziz, Q. Zhou, U.R. Desai, Rational design of potent, small, synthetic allosteric inhibitors of thrombin, J. Med. Chem. 54 (2011) 5522–5531, https://doi.org/10.1021/jm2005767.

[51] H. Shukla, V. Kumar, A.K. Singh, S. Rastogi, S.R. Khan, M.I. Siddiqi, M.Y. Krishnan, M.S. Akhtar, Isocitrate lyase of Mycobacterium tuberculosis is inhibited by quercetin through binding at N-terminus, Int. J. Biol. Macromol. 78 (2015) 137–141, https://doi.org/10.1016/j.ijbiomac.2015.04.005.

[52] H.-L. Cheah, V. Lim, D. Sandai, Inhibitors of the Glyoxylate Cycle Enzyme ICL1 in Candida albicans for potential use as antifungal agents, PLoS ONE 9 (2014), e95951. https://doi.org/10.1371/journal.pone.0095951.

[53] B. Suriyanarayanan, K. Shanmugam, R. Sarojini Santhosh, Synthetic quercetin inhibits mycobacterial growth possibly by interacting with DNA gyrase, Romanian Biotechnological Letters. 18 (2013) 8587-8593. <https://www.rombio.eu/vol18nr5/4 R.S.pdf> (accessed July 18, 2019).

[54] D. Butov, S. Zaitseva, T. Butova, G. Stepanenko, O. Pogorelova, N. Zhelezniakova, Efficacy and safety of quercetin and polyvinylpyrrolidone in treatment of patients with newly diagnosed destructive pulmonary tuberculosis in comparison with standard antimycobacterial therapy, Int. J. Mycobacteriol. 5 (2016) 446–453, https://doi.org/10.1016/j.ijmyco.2016.06.012.